

BA2: Digital Korea

**Week 5: Practice & Deepen
From Preprocessing to Description**

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Today's Agenda

1. Research in action: CTA and Korean Studies
2. Review: Core concepts
3. R programming check-in
4. Descriptive exploration tools in Orange
5. Break
6. Hands-on: Exploring presidential speeches in Orange
7. Looking ahead: Week 6 assessment

Research in Action

CTA in Korean Studies

Your corpus, your questions

Research questions you can ask

- Which presidents talked most about 경제?
- When did 통일 disappear?
- How did rhetoric shift across 30 years?

The tools you already have

- A preprocessed Korean corpus
- Bag of Words / TF-IDF
- Word clouds, bar charts, metadata

Today's goal

Learn to **explore descriptively** — the step between preprocessing and deeper analysis.

Review: Core Concepts

Key Terms Review

Make sure you can explain each of these

Term	Definition	Description	Think of it as...
BoW	Bag of Words	Represent text by word counts, ignoring order and grammar	Cutting words out, dropping them in a bag
TF	Term Frequency	Count of a word in one document (can be normalized by length)	One cell of the matrix
DF	Document Frequency	Number of documents in which a word appears	One column's spread across rows
DTM	Document-Term Matrix	The table (docs $\times$ words) that BoW creates	The spreadsheet the computer "sees"
IDF	Inverse Doc. Freq.	$\log(N/DF)$; how rare a word is across the corpus	High IDF = rare = informative
TF-IDF	$TF \times IDF$	Weighting that highlights words frequent here but rare across the corpus	"Frequent here, rare elsewhere"

Matrix vs. DTM vs. Dataframe

Matrix

A grid of numbers. No labels, no column names — just rows and columns of values.

General math concept.

DTM

A *specific* matrix: rows = documents, columns = terms, cells = counts (or TF-IDF weights).

What BoW creates.

Dataframe

A table with named columns of mixed types (text, numbers, dates).

What you see in the Corpus widget.

Your CSV is this.

Key distinction

Your CSV is a **dataframe**. After Bag of Words, it becomes a **DTM**. Both are tables — but the DTM is all numbers, built for computation.

What the BoW Widget Actually Does

The Bag of Words widget builds a DTM and gives you three options that change the cell values:

1. **Count** — each cell is the raw integer count of a word in a document
2. **IDF toggle** — multiplies each cell by the word's IDF weight. Common words get down-weighted (e.g., 국민: $IDF \approx 0.19$)
3. **Regularization** (= L2 normalization) — scales each document row to unit length, removing the effect of document length differences

Why the Word Cloud shows decimals

The Word Cloud displays the **mean value across all documents** for each word. With Count, a word with 3,977 total occurrences across 749 speeches shows as ≈ 5.31 . Toggle IDF on and that same word might drop to ≈ 1 because its high frequency is down-weighted.

Self-Check

Check your understanding with these review questions:

1. What information does the Bag of Words model **discard**?
2. If a word has high TF but low TF-IDF, what does that tell you?
3. What is the difference between TF and DF?
4. Why is the DTM mostly zeros (sparse)?
5. A word appears in 748 out of 749 speeches. Is its IDF high or low?

R Programming Check-In

DataCamp Check-In

Introduction to R: Chapters 4 (Factors) & 5 (Matrices)

Current assignment: Factors & Matrices — extended deadline **March 9**.

- Factors: categorical data, levels, ordering
- Matrices: 2D structures, row/column operations
- These connect directly to the DTM concept

Extra credit opportunity

Complete **all supplementary DataCamp exercises** (beyond required chapters) → **+0.25** on your final assignment grade.

Any questions or difficulties with R so far?

Descriptive Exploration in Orange

Why Start with Descriptive Analysis?

Before you dive deeper, explore

- Validates your preprocessing
- Reveals corpus structure
- Generates hypotheses
- Catches data quality issues early

What it gives you

- Most frequent terms overall
- Distinctive terms per group
- Distribution patterns
- Keywords in context

Descriptive analysis is not optional

Skipping exploration means you won't notice problems until later — wrong column, leftover stopwords, unexpected patterns. Always look at your data first.

Our Toolkit

Widget	What it shows	When to use it
Word Cloud	Visual overview of frequent terms	First look at your data
Bar Plot	Precise term frequencies or TF-IDF scores	Comparing values across terms
Concordance	Keywords in their original context	Understanding word usage
Statistics	Per-document metrics (word count, etc.)	Spotting outliers, data quality
Select Rows	Filters corpus by metadata	Creating subsets (by president, year)

Statistics Widget

The **Statistics** widget reports per-document metrics:

- **Word count** per document
- **Unique words** per document
- Useful for spotting **outliers**: unusually short or long documents
- Helps identify **data quality issues**: empty documents, duplicates

What to look for

If one speech has 50 words and another has 5,000, that difference will affect your analysis. The Statistics widget makes this visible *before* you start interpreting results.

Word Clouds: What They Show (and Don't)

What you can read

- Most frequent terms at a glance
- Relative size = relative frequency
- Quick preprocessing sanity check
- Color by metadata variable

Limitations

- Imprecise — size differences are hard to judge
- No exact counts visible
- Layout is random each time
- Can be misleading for comparison

Rule of thumb

Word clouds are useful for a **first look**. Always follow up with bar charts for anything you want to claim or compare precisely.

Bar Charts: Precise Comparison

The **Bar Plot** widget gives you what word clouds cannot: exact values and clear ranking.

- Both Word Cloud and Bar Plot display the **mean value per word** from the DTM
- What those values *are* depends on your BoW settings:
 - **Count** → mean raw frequency across documents
 - **Count + IDF** → mean IDF-weighted frequency
- **Color by metadata** (e.g., president) for comparison

Try this

Switch your BoW widget between Count and Count + IDF — watch how the rankings change. Terms that rank high under Count but drop with IDF are common across the corpus (high DF), not distinctive.

Concordance: Keywords in Context

The **Concordance** widget shows every occurrence of a word with its surrounding text — a KWIC (keyword in context) display.

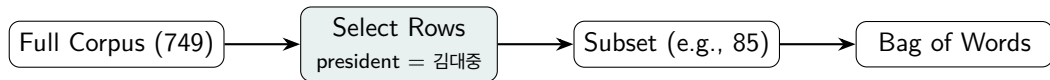
Left context	Keyword	Right context
...평화와	통일	을 위해 노력...
...남북 경제	통일	시대를 준비...
...기본질서에 의한	통일	정책을 추진...

Bridge between quantitative and qualitative

A word count tells you *how often*. Concordance tells you *how* — same word, different meanings across presidents and decades.

Subsetting: Select Rows

The **Select Rows** widget filters your corpus by metadata values.



- Filter by president, year, speech type, or any metadata column
- Place **before** Bag of Words to focus your analysis on one group
- Connect visualization widgets to the BoW output for focused exploration

New Corpus: Korean History Textbooks

NIKH (국사편찬위원회) History Textbook Corpus — demo subset

Era	Books	Period covered
Colonial	3	Japanese colonial rule (1910–1945)
Authoritarian	3	Rhee–Park–Chun (1954–1987)
Democratic	3	6th–7th Curriculum (1995–2002)

- 9 textbooks, ~1.8 MB — loads quickly in Orange
- Key metadata: era, period, level, year
- Text column: `full_text` (raw) → `processed_text` (after script)

Research angle

How did the teaching of Korean history change across political eras?

Demo

Exploring Korean history textbooks in Orange

Break

We will resume in 10 minutes.

Hands-On Activity

Hands-On: Exploring History Textbooks

Corpus: 9 Korean history textbooks from 3 eras (Colonial, Authoritarian, Democratic).

1. Load the CSV → **File** → **Corpus** (select `full_text`)
2. Run the **Python Script** (preprocessing) → **Corpus** (select `processed_text`)
3. Add **Select Rows** → filter by era
4. Connect to **Bag of Words** → start with Count
5. Add **Word Cloud** and **Bar Plot** → explore top terms
6. Switch the era in Select Rows → compare

Work in pairs

Follow the same workflow we used for presidential speeches — just a new corpus.

Guiding Questions

1. What are the **top terms** for each era? Do they match what you'd expect?
2. How do rankings change when you switch from **Count** to **TF-IDF**?
3. Compare Colonial vs. Democratic — which terms are **shared** vs. **unique**?

If you finish early:

- Use **Concordance** — pick a term and see how its context differs across eras
- Try enabling **Regularization** (L2 normalization) in BoW — how does it affect results?
- Can you find a term that ranks high under Count but drops under TF-IDF? Why?

Debrief

Let's discuss as a class:

1. What was the most **surprising** finding from your exploration?
2. Did switching from Count to TF-IDF change your interpretation?
3. What **limitations** did you notice in this approach?
4. How might you **follow up** on what you found? What additional analysis would help?

Remember

Descriptive exploration generates **hypotheses**. The methods we learn in Weeks 7–10 will help you **test** them.

Looking Ahead

Week 6: Assessment Preview

Part 1: Online Quiz (~20 min)

- Multiple-choice questions covering Weeks 1–5 concepts

Part 2: Hands-On Task (~20 min)

- Download a small new corpus, load it into Orange, and preprocess it
- Produce a clean Word Cloud showing successful preprocessing
- Upload to your shared private GitHub repo:
 - Your .ows file and a saved image of the Word Cloud
 - A short .md file: what research question might you ask with this data, and what would you expect to find?

How to prepare

A **study guide** is posted on the Assignments tab of the course website (Week 5). Focus on reviewing concepts and practicing in Orange.

Closing

This week:

- Review all key concepts (study guide on course website)
- Practice building workflows in Orange
- Complete DataCamp Factors & Matrices (due Mar. 9)

Thank you!

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